SYNOPSIS REPORT

On

Mera Godaam: Inventory Management System

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF BUSINESS ADMINISTRATION (COMPUTER APPLICATION)



Submitted by

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ARIHANT COLLEGE OF ARTS, COMMERCE AND SCIENCE, CAMP, PUNE
FEB 2025

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CERTIFICATE

This is to certify that the Project Report entitled **Mera Godaam: Inventory Management System** prepared by **Suraj Uttamchand Oswal** students of Bachelor of Business Administration (Computer Application) semester-V course at Arihant College of Art's, Commerce and Science(ACACS), Pune-01.

To the best of our knowledge, this is original study done by the said student and important sources used by them have been duly acknowledged in this report.

The report is submitted in partial fulfilment BBA(CA) Semester-V syllabus as per rule prescribed guideline of Savitribai Phule Pune University.

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Project Guide	**HOD**	**I/C Principal**

Internal Examiner **External Examiner**

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❖ Name of Project:

Mera Godaam: Inventory Management System

❖ Project Partner Name and Roll No:

Roll No.	Name.
70	Suraj Oswal

❖ Introduction

The project titled "**Mera Godaam: Inventory Management System for Smart General Store"** is developed as part of my academic coursework to understand the concepts of database management, server-side programming, and web application development. The main objective of this project is to design and implement a basic solution that helps a store efficiently manage its daily sales, stock levels, customer records, and purchase orders.

In many small and medium stores, inventory management is often done manually using registers or spreadsheets, which leads to mistakes such as miscounting stock, loss of records, and slow billing. To overcome these problems, I have developed a **web-based inventory management system** that automates these tasks.

This system allows the **store manager** to:

- Add, update, and delete items in the inventory.
- Record daily sales and automatically reduce stock quantity after each sale.
- Generate professional PDF bills for customers with all billing details.
- Track today's total sales and maintain a daily history of transactions.
- Identify items with low or zero quantity and generate purchase orders for refilling from suppliers.

The project follows an object-oriented approach, and the database design ensures data consistency and integrity. Technologies used in the system include HTML, CSS, JavaScript (frontend), Node.js and Express.js (backend), and MySQL (database). The system is structured as a single-page application (SPA), where all pages load dynamically within one main interface for better performance and user experience.

Through this project, I have learned how to connect frontend and backend technologies, handle real-time data, generate dynamic reports in PDF format, and design a logical workflow for a real-world business problem. It demonstrates my understanding of both programming and database concepts and provides a practical example of how modern technologies can simplify daily store operations.

❖ Existing System

In the existing system, most inventory and billing processes are carried out manually or through basic spreadsheet tools. Store managers usually record item details, supplier information, and sales transactions in physical registers or simple Excel sheets. This approach makes it difficult to manage stock levels accurately, track sales, and generate reports efficiently.

When items go out of stock, the store manager has to manually verify available quantities and contact suppliers individually, which leads to delays and errors. Billing is also handled manually, where totals are calculated using calculators and handwritten receipts are given to customers. This process is time-consuming and increases the chances of mistakes in calculations and data entry.

Moreover, the existing system lacks sales, purchase orders, and inventory data are not automatically updated in one place. This causes mismatched records, duplication of data, and confusion in day-to-day operations. Generating reports or analyzing sales trends becomes a lengthy and mistake in task.

Therefore, an automated inventory management system is required to streamline these operations, reduce manual effort, and improve accuracy and efficiency in managing store activities.

❖ Proposed System

The proposed system aims to automate the overall process of inventory management and billing in a retail store. It replaces the traditional manual approach with a computerized system that efficiently manages items, sales, purchases, and suppliers within a single platform.

In this system, the store manager can easily **add, update, and delete items** from the inventory with complete details such as item name, SKU, category, quantity, price, supplier name, and description. When an item is sold, its quantity is automatically reduced from stock, ensuring real-time inventory tracking.

The system also includes a **billing module**, which allows the user to generate bills quickly. Each bill automatically includes customer details, sold items, quantity, price, discount, and total amount. The system can also generate a **PDF receipt** for each sale, making it easier to maintain digital records.

In addition, the proposed system helps manage **low stock and purchase orders**. When the stock of certain items drops below a specific level, the system can identify the supplier's name and allow the manager to reorder items directly. This ensures the store always maintains sufficient stock.

- The system provides the following benefits:
- Faster and more accurate billing process
- Automatic stock updates after each sale
- Easy tracking of suppliers and items
- Secure storage of sales and purchase data
- Daily sales summary and reporting features

Overall, the proposed system reduces mistakes, saves time, and improves the efficiency of store operations by integrating all inventory and billing functions into a single, user-friendly application.

❖ Scope

The scope of this project is to design and develop a **computerized inventory management and billing system** that helps retail stores manage their stock, sales, and purchases in a more organized and efficient way. The system will automate the key processes of item management, billing, and supplier handling, reducing the need for manual record-keeping.

The system allows the **store manager or owner** to perform several important tasks such as adding new items, updating existing stock, generating sales bills, and managing low stock. It also maintains detailed information about each sale, including customer details and payment methods, ensuring complete transaction records.

The system will be developed using **HTML, CSS, JavaScript, Node.js, and MySQL,** making it simple, lightweight, and easy to use on local or small business setups. It can later be expanded with additional features such as user authentication, analytics dashboards, and online access for better scalability.

Overall, the project provides a reliable and user-friendly solution for managing daily store operations while minimizing errors and improving business efficiency.

❖ Objectives

- **To automate the inventory process** and remove manual mistakes in tracking items.
- **To maintain accurate records** of stock, sales, customers, and suppliers.
- **To generate bills instantly** with complete customer and item details.
- **To update stock automatically** after every sale.
- **To identify low stock items** and do purchase orders for refilling.
- **To generate a printable PDF bill** for each transaction.
- **To provide daily sales summaries** for better monitoring and business decisions.
- **To create an easy-to-use interface** suitable for small store managers with basic computer knowledge.

❖ Modules

- **Item Management Module** - Handles adding, updating, deleting, and viewing items. It maintains stock quantity and item details such as name, price, category, and supplier.
- **Sales & Billing Module** - Manages sales transactions, applies discounts, calculates totals, and generates PDF bills with customer and item details.
- **Customer Management Module** - Stores and manages customer information like ID, name, and contact number for tracking and reporting purposes.
- **Supplier Management Module** - Keeps records of suppliers and their items, helping in restocking and purchase order generation.
- **Purchase Order Module** - Used for refilling low-stock items by creating purchase orders linked to specific suppliers.
- **Report Module** - Displays daily sales and summary reports for quick performance analysis.

❖ Software Requirements

- **Operating System**: Windows 10 or above
- **Frontend Technologies**: HTML, CSS, JavaScript
- **Backend Technology**: Node.js with Express.js
- **Database**: MySQL
- **Tools & IDEs**: Visual Studio Code, MySQL Workbench
- **Browser**: Google Chrome or Microsoft Edge
- **Libraries & Modules**: Express, Multer, CORS, PDFKit, MySQL2

❖ Hardware Requirements

- **Processor**: Intel Core i3 (6th Generation) or higher
- **RAM**: Minimum 8 GB
- **Storage**: 256 GB SSD or higher
- **Display**: 1366×768 resolution or above
- **Other Devices**: Keyboard, Mouse, and Printer (for PDF bill printing)

❖ Merits

- The system simplifies inventory management by automating item tracking and billing.
- It reduces mistakes that usually occur during manual entry or calculation.
- Generates professional PDF bills automatically after each sale.
- Keeps record of sales, customers, and suppliers efficiently in the database.
- Allows real-time updates in item quantity after every sale.
- Provides a clear view of low stock items and supplier details for easy reordering.
- Improves overall store management and saves time compared to paper based systems.

❖ Limitations / Demerits

- The system currently works on a single local server and not online (no cloud integration).
- Requires basic technical knowledge to operate and maintain the Node.js server.
- Lacks advanced analytics or reporting features.
- Does not support multi user access or login system.
- Internet connection may be needed for future online synchronization or updates.

❖ Application

- **General Stores** - to record sales, manage product stock, and generate bills.
- **Supermarkets** - to monitor product quantities and automate reordering when items are low.
- **Pharmacies** - to manage medicine stock, track expiry dates, and maintain purchase details.
- **Electronics or Clothing Shops** - to handle multiple products, prices, and supplier details.
- **Wholesale or Retail Businesses** - to maintain customer records and generate invoices easily.

❖ Feasibility Study

1. Technical Feasibility

The project is technically feasible because it uses commonly available and easy-to-learn technologies such as HTML, CSS, JavaScript, Node.js, Express.js, and MySQL.

These technologies can run on a normal computer without needing high-end hardware. The system was successfully developed and tested using a Windows 10 laptop with 8GB RAM and Intel i3 processor, which proves that the project can be implemented with limited technical resources.

2. Economic Feasibility

The system is economically feasible since it does not require any paid tools or licenses. All software components like Visual Studio Code, Node.js, and MySQL Workbench are open-source and free to use.

As a student project, the development cost is almost zero - only time and effort were invested. This makes it a cost-effective solution for small stores or startups that want a simple inventory management system without heavy investment.

3. Operational Feasibility

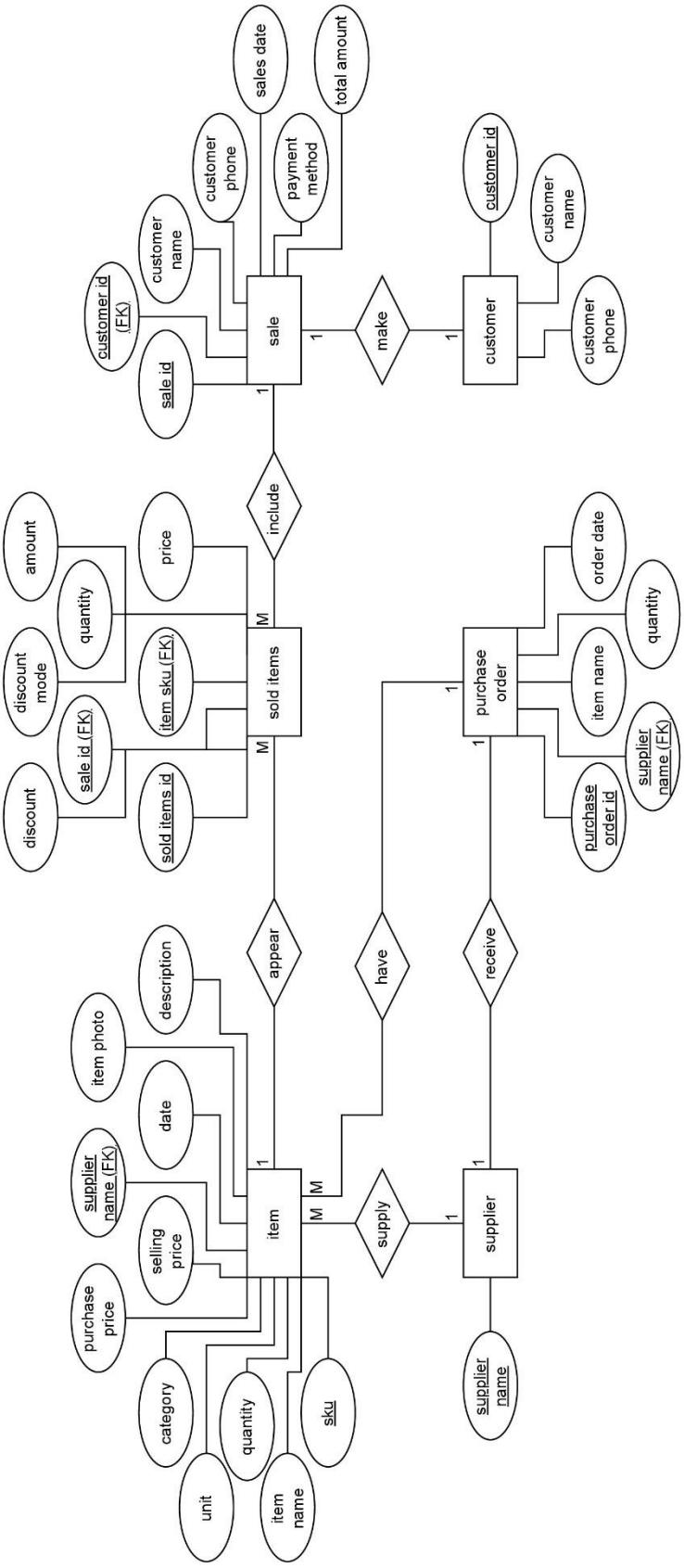
The system is easy to operate and user-friendly. Store managers can easily perform operations like adding items, updating stock, recording sales, and generating bills without needing deep technical skills.

The user interface is clean and simple, designed to resemble real world store operations. It reduces manual effort and helps in keeping accurate records, making it practical and feasible for day-to-day use in small or medium general stores.

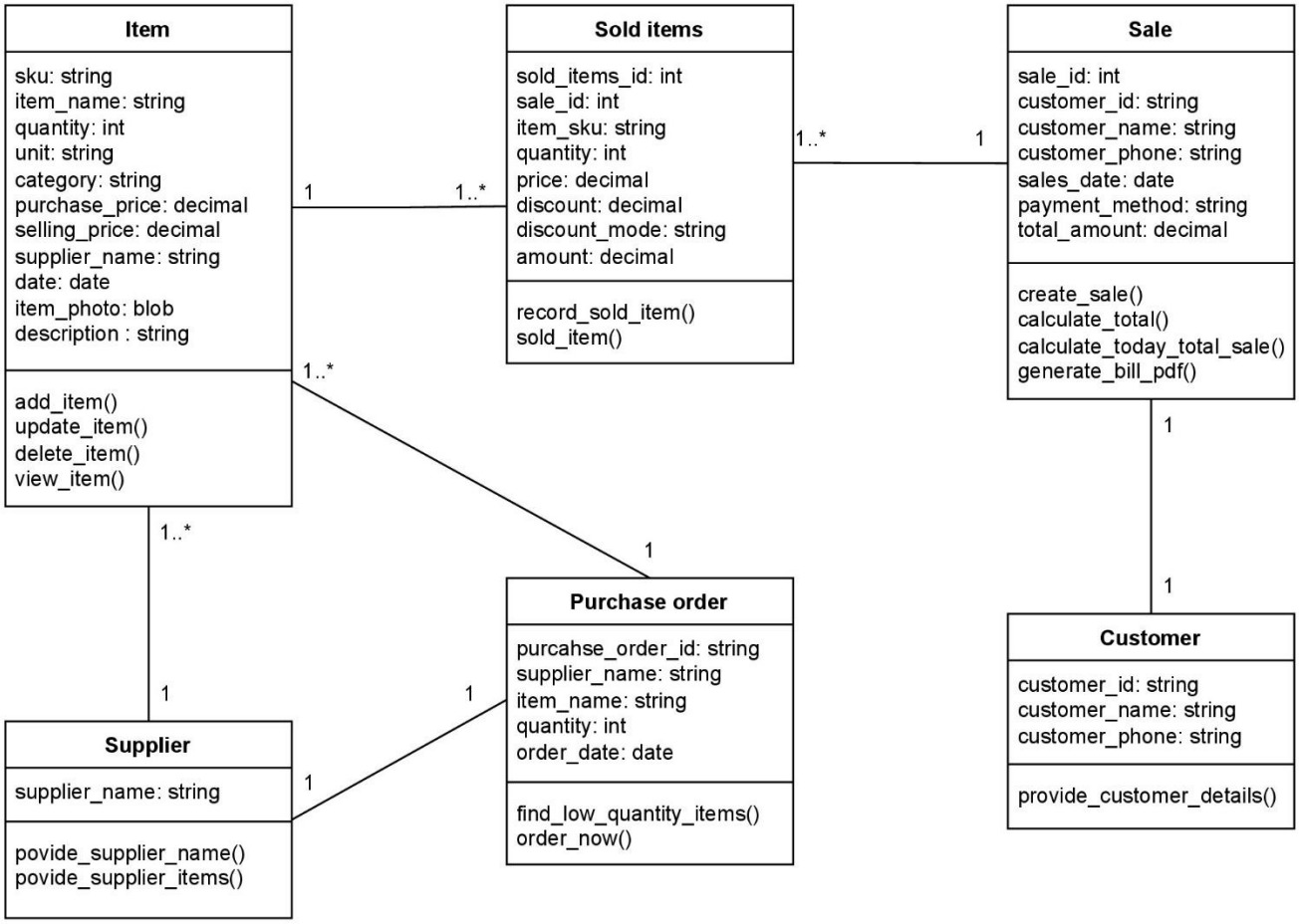
❖ Future Enhancements

- **Barcode Integration**: To speed up the billing and stock update process by scanning item barcodes.
- **Email or SMS Notifications**: To alert the store manager or supplier when an item's stock is low or an order is confirmed.
- **Multi-User Access**: Allowing multiple employees to log in with different access levels (e.g., admin, cashier, manager).
- **Data Analytics**: To generate sales reports, profit analysis, and inventory trends for better decision-making.
- **Cloud Storage**: To store all records online so that the system can be accessed from anywhere.
- **Mobile Application**: Developing a mobile version of the system for easier management on smartphones.

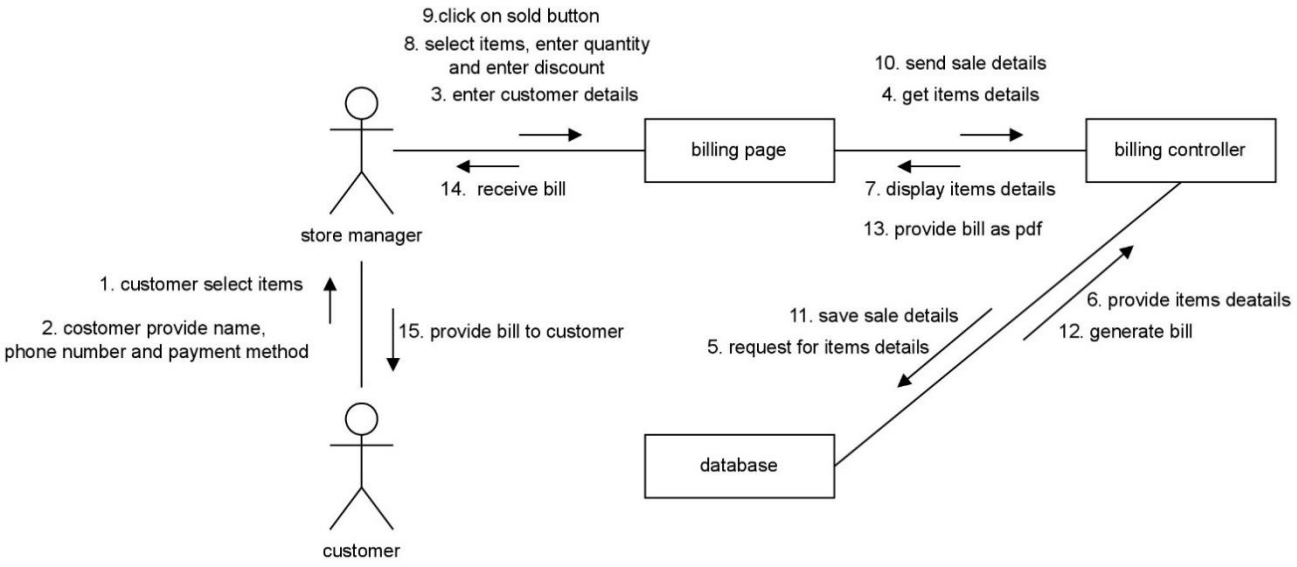
❖ E-R Diagram Of Inventory Management System



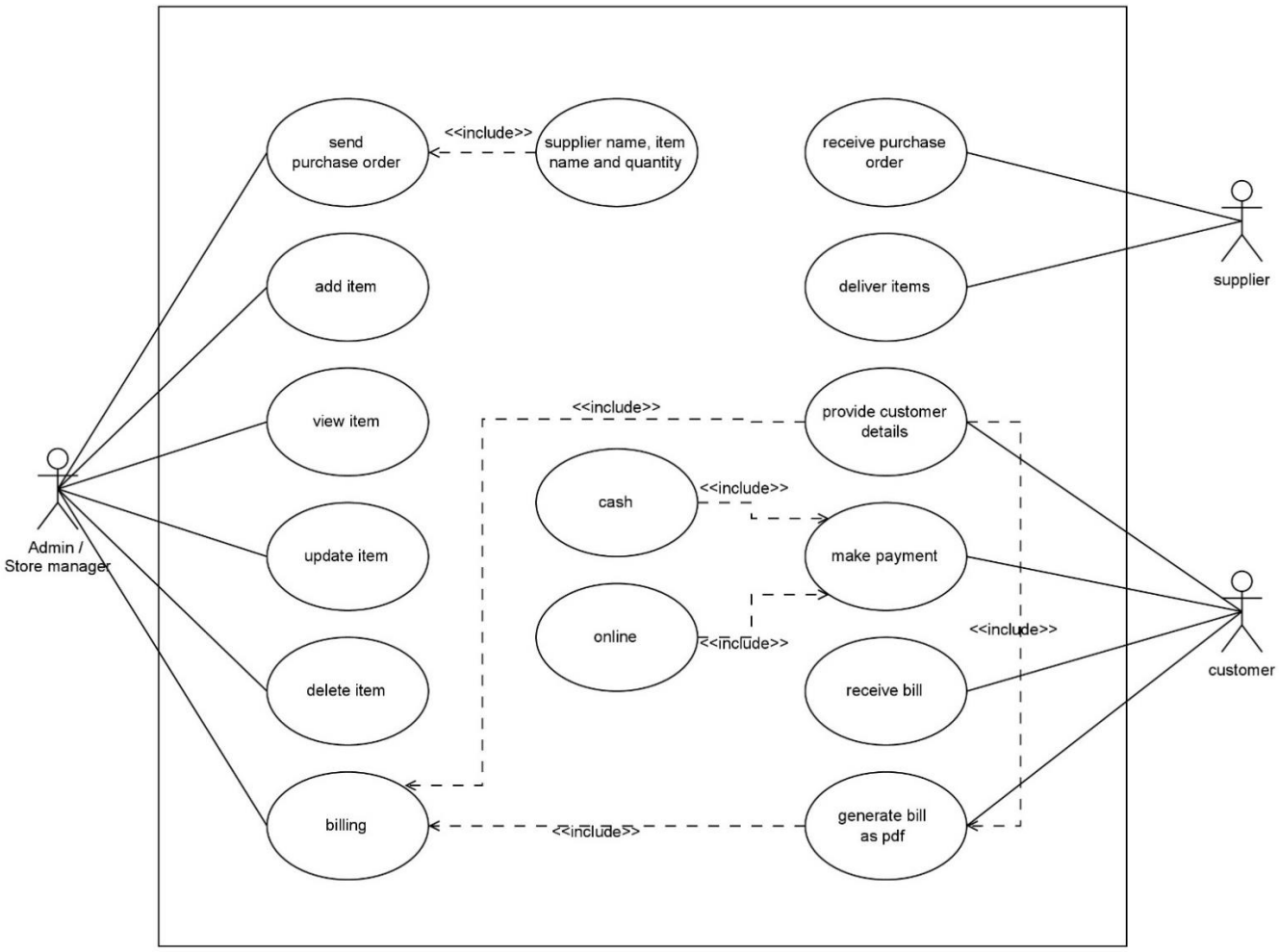
❖ Class Diagram Of Inventory Management System



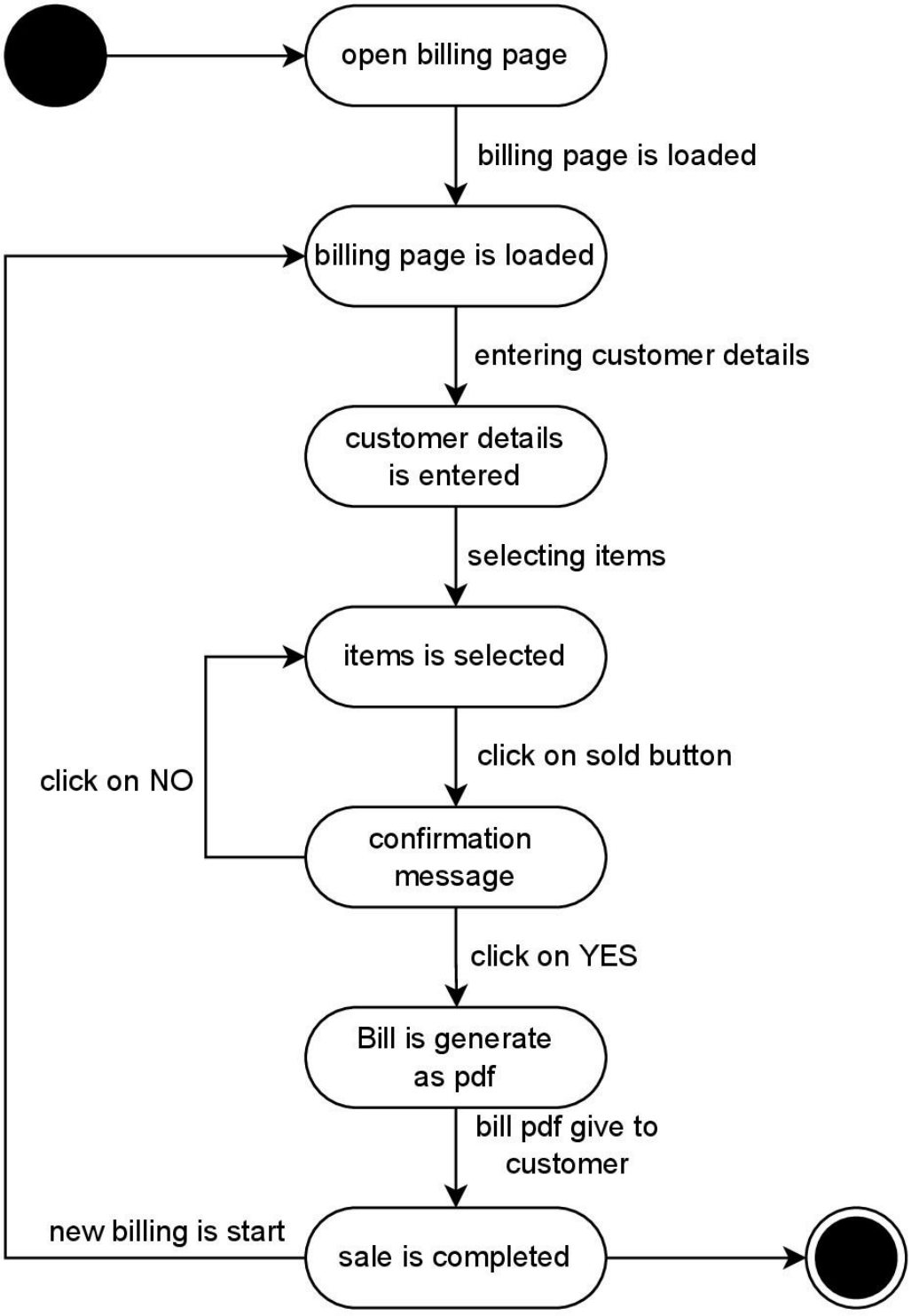
❖ Collaboration Diagram Of Inventory Management System



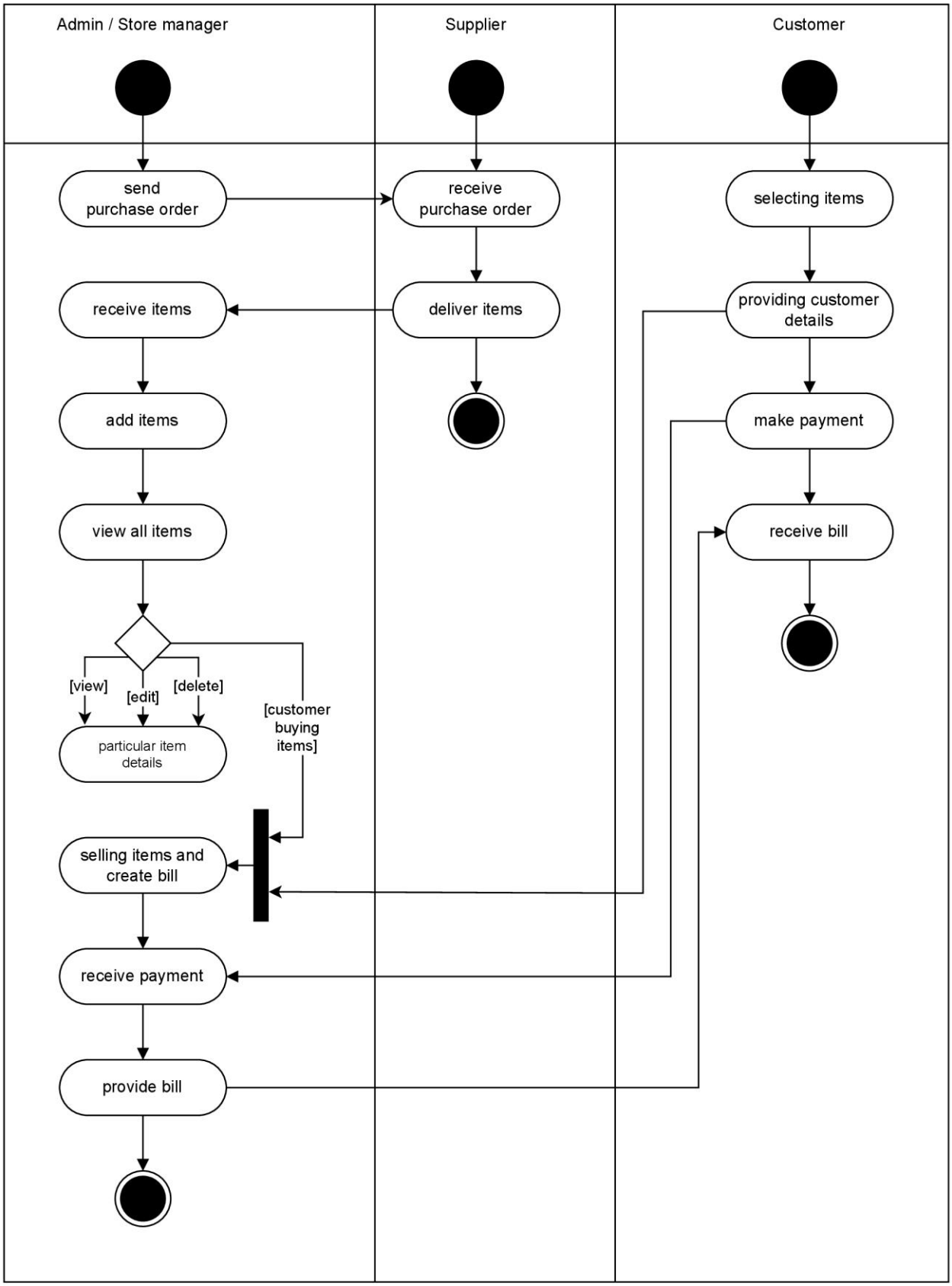
❖ Use Case Diagram Of Inventory Management System



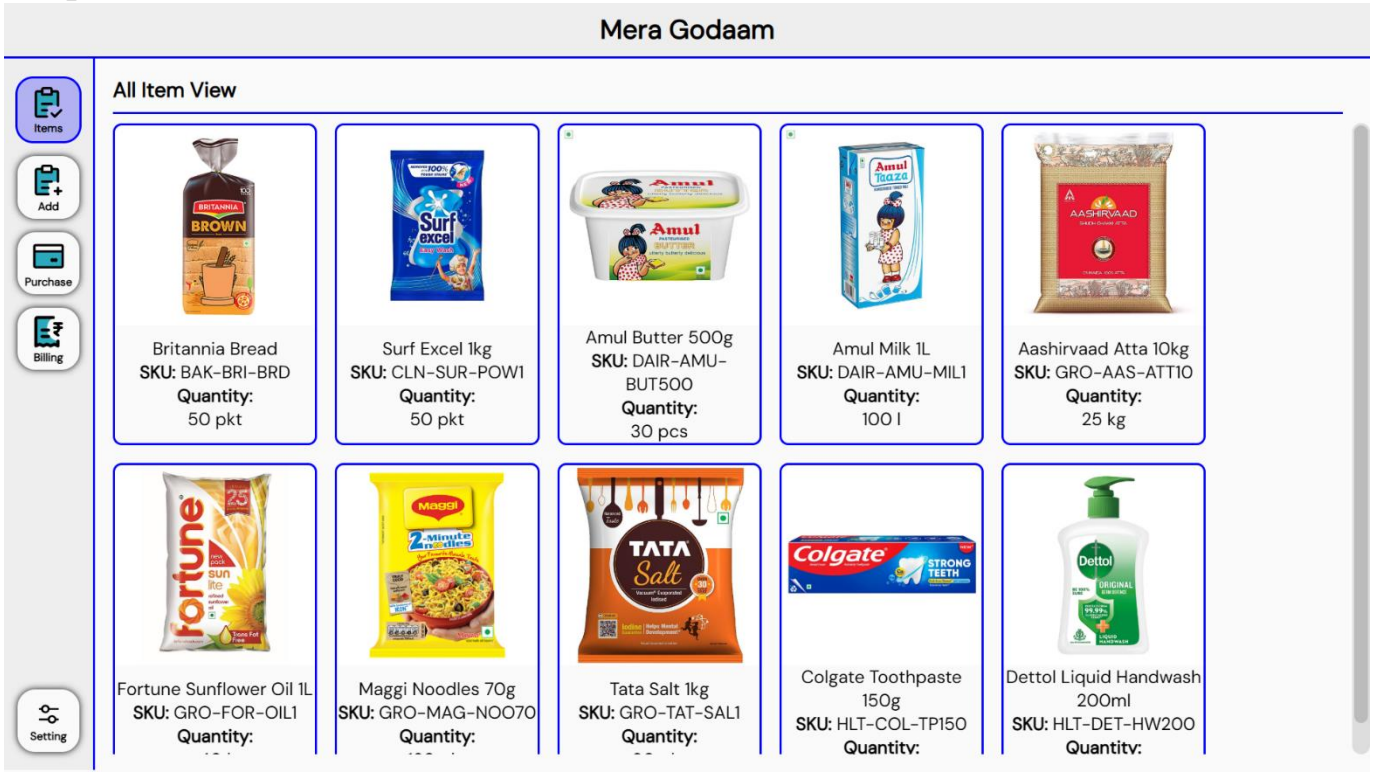
❖ **State Chart Diagram Of Inventory Management System**

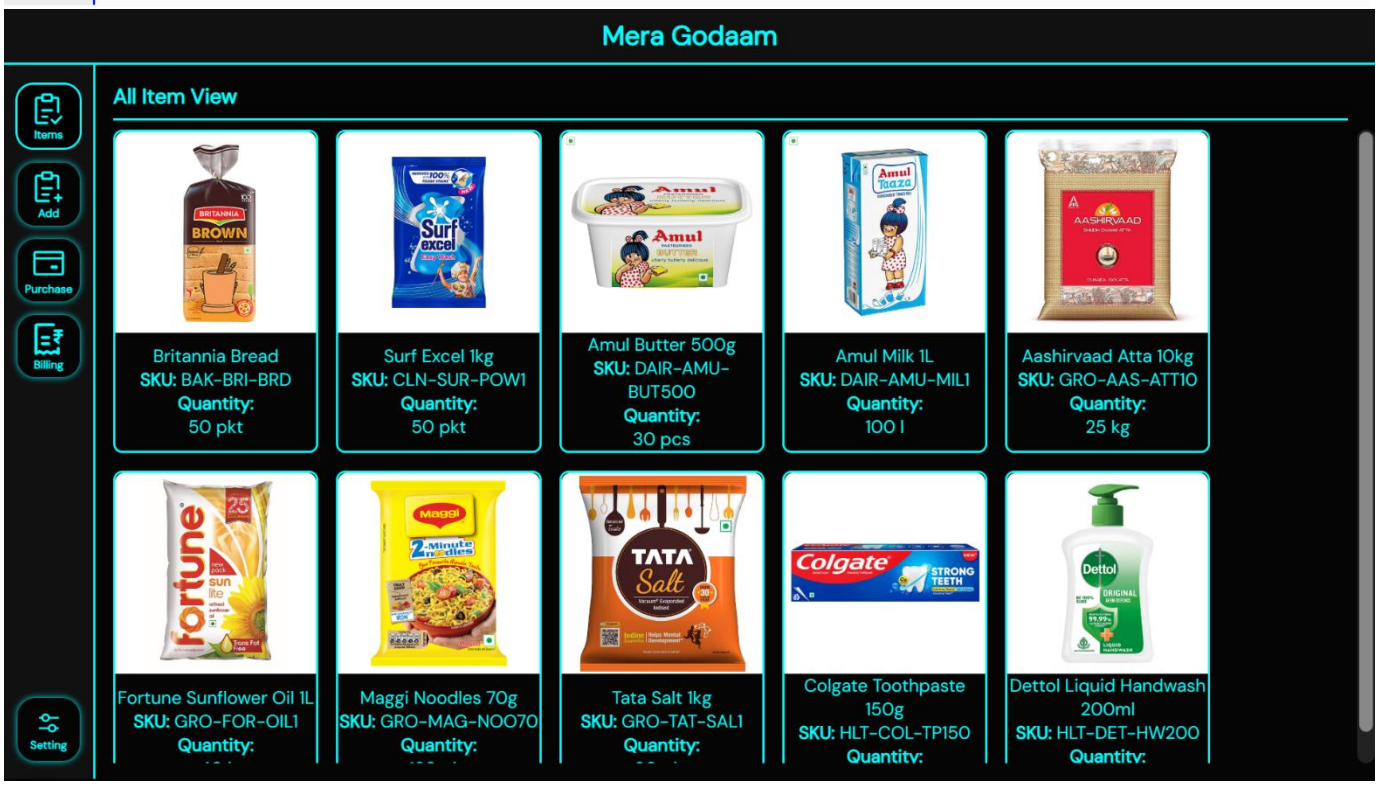


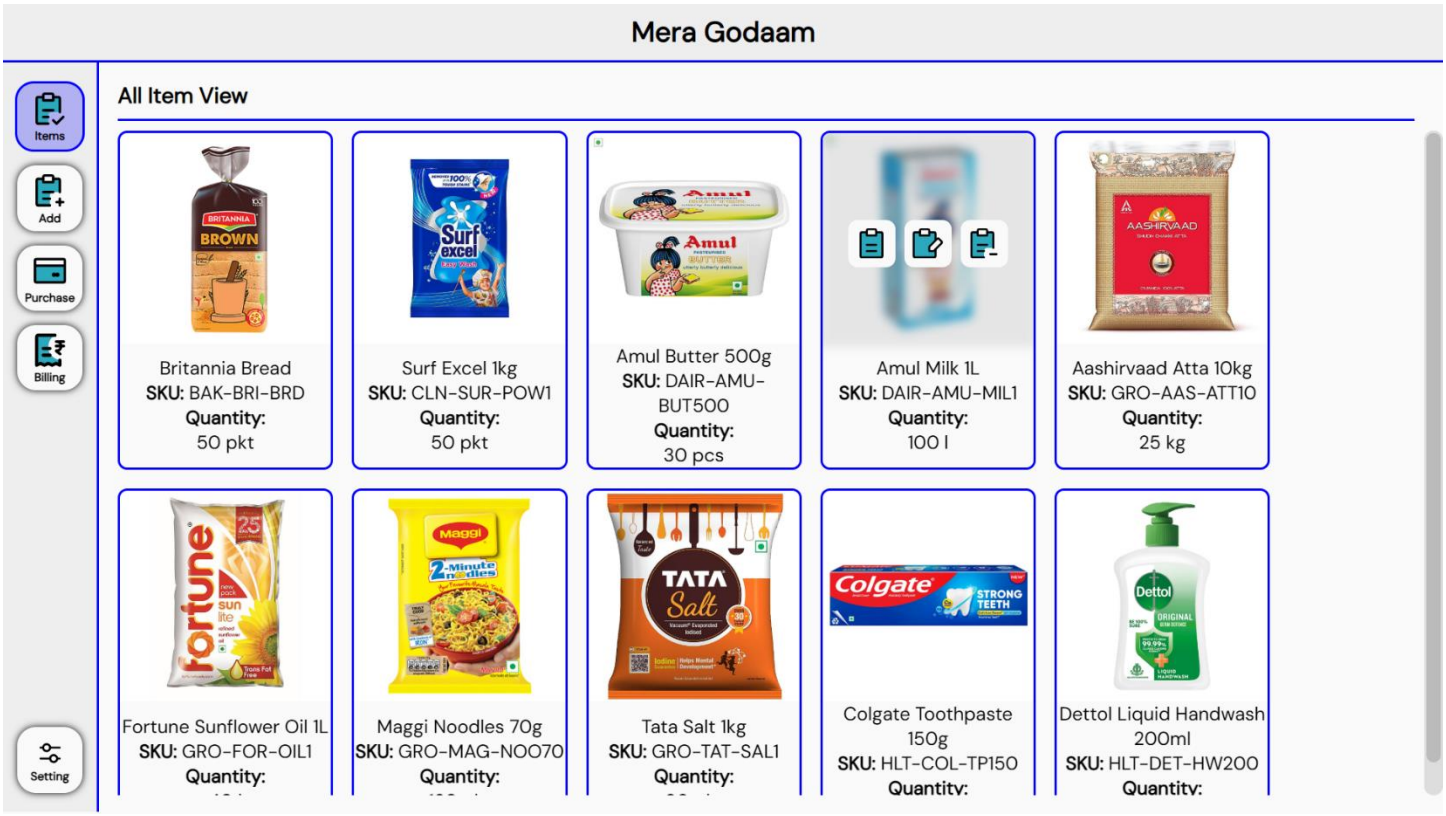
❖ Activity Diagram Of Inventory Management System

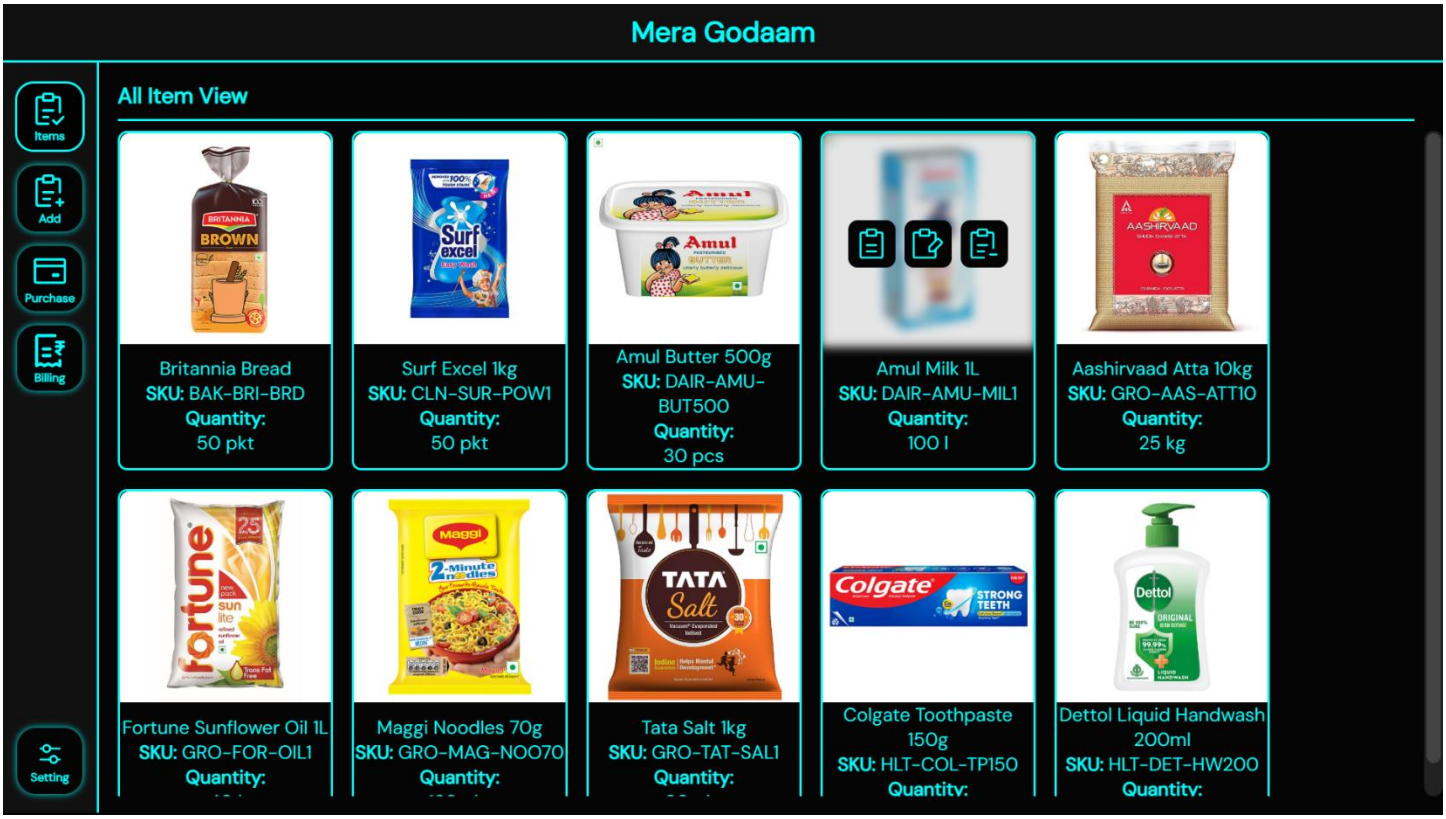


❖ Output Screenshots






Mera Godaam
Items
Add
Purchase
Billing
Setting
All Item View
Britannia Bread
SKU: BAK-BRI-BRD
Quantity:
50 pkt
Surf Excel 1kg
SKU: CLN-SUR-POW1
Quantity:
50 pkt
Amul Butter 500g
SKU: DAIR-AMU-BUT500
Quantity:
30 pcs
Amul Milk 1L
SKU: DAIR-AMU-MIL1
Quantity:
100 l
Aashirvaad Atta 10kg
SKU: GRO-AAS-ATT10
Quantity:
25 kg
Fortune Sunflower Oil 1L
SKU: GRO-FOR-OIL1
Quantity:
Maggi Noodles 70g
SKU: GRO-MAG-NOO70
Quantity:
Tata Salt 1kg
SKU: GRO-TAT-SAL1
Quantity:
Colgate Toothpaste 150g
SKU: HLT-COL-TP150
Quantity:
Dettol Liquid Handwash 200ml
SKU: HLT-DET-HW200
Quantity:


Mera Godaam
Items
Add
Purchase
Billing
Setting
All Item View
Britannia Bread
SKU: BAK-BRI-BRD
Quantity:
50 pkt
Surf Excel 1kg
SKU: CLN-SUR-POW1
Quantity:
50 pkt
Amul Butter 500g
SKU: DAIR-AMU-BUT500
Quantity:
30 pcs
Amul Milk 1L
SKU: DAIR-AMU-MIL1
Quantity:
100 l
Aashirvaad Atta 10kg
SKU: GRO-AAS-ATT10
Quantity:
25 kg
Fortune Sunflower Oil 1L
SKU: GRO-FOR-OIL1
Quantity:
Maggi Noodles 70g
SKU: GRO-MAG-NOO70
Quantity:
Tata Salt 1kg
SKU: GRO-TAT-SAL1
Quantity:
Colgate Toothpaste 150g
SKU: HLT-COL-TP150
Quantity:
Dettol Liquid Handwash 200ml
SKU: HLT-DET-HW200
Quantity:

Mera Godaam

Items
Add
Purchase
Billing
Setting

Add New Item

Item Name: Amul Milk 1L

SKU: DAIR-AMU-MIL1

Quantity: 100

Unit: Liter (l)

Category: Dairy & Eggs

Selling Price: Rs. 64.00

Purchase Price: Rs. 58.00

Supplier Name: Amul Dairy

Date Added: 12-10-2025



Description: Amul Milk 1L

Reset | Update | Delete

Mera Godaam

Items
Add
Purchase
Billing
Setting

Add New Item

Item Name: Amul Milk 1L

SKU: DAIR-AMU-MIL1

Quantity: 100

Unit: Liter (l)

Category: Dairy & Eggs

Selling Price: Rs. 64.00

Purchase Price: Rs. 58.00

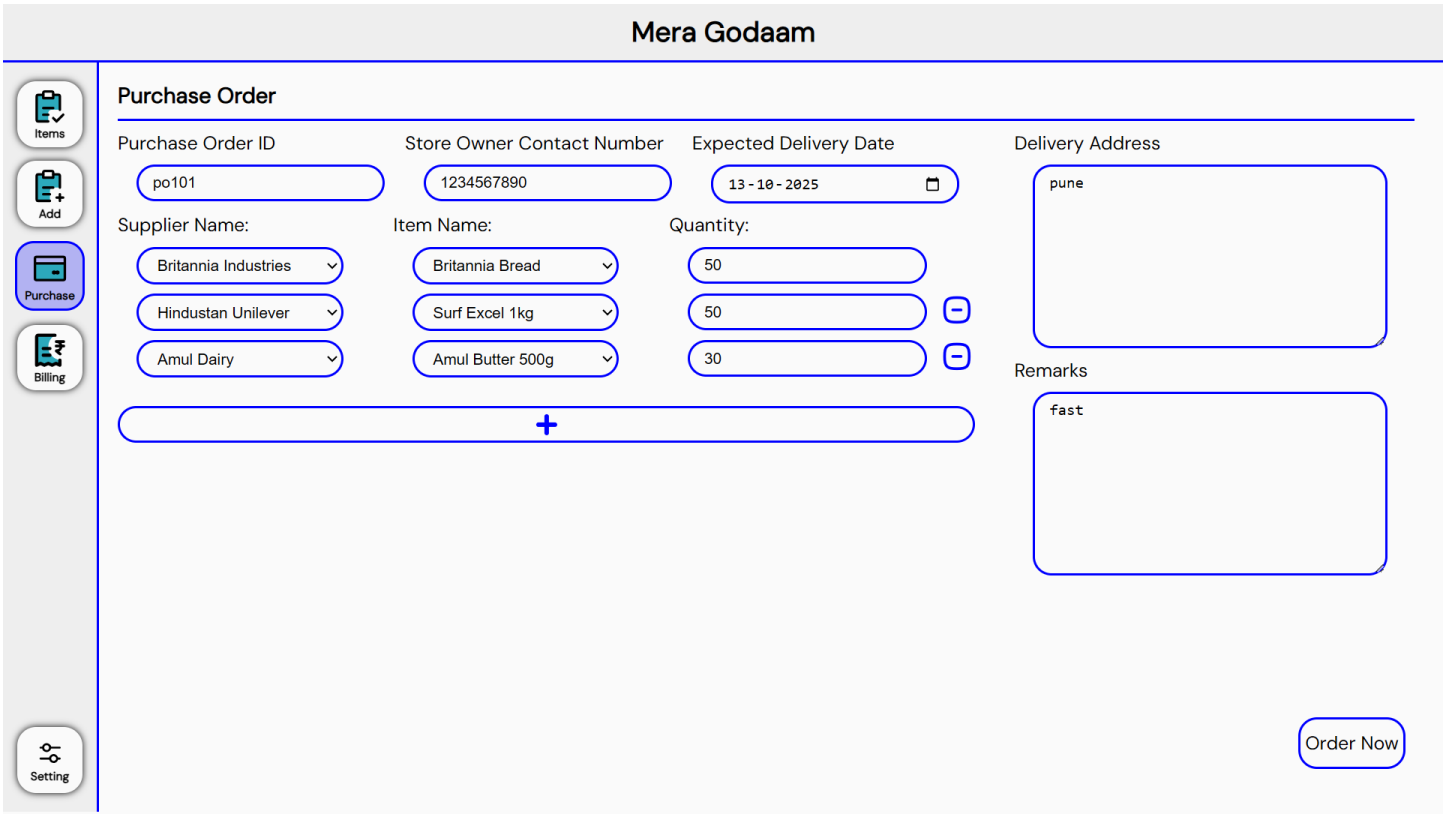
Supplier Name: Amul Dairy

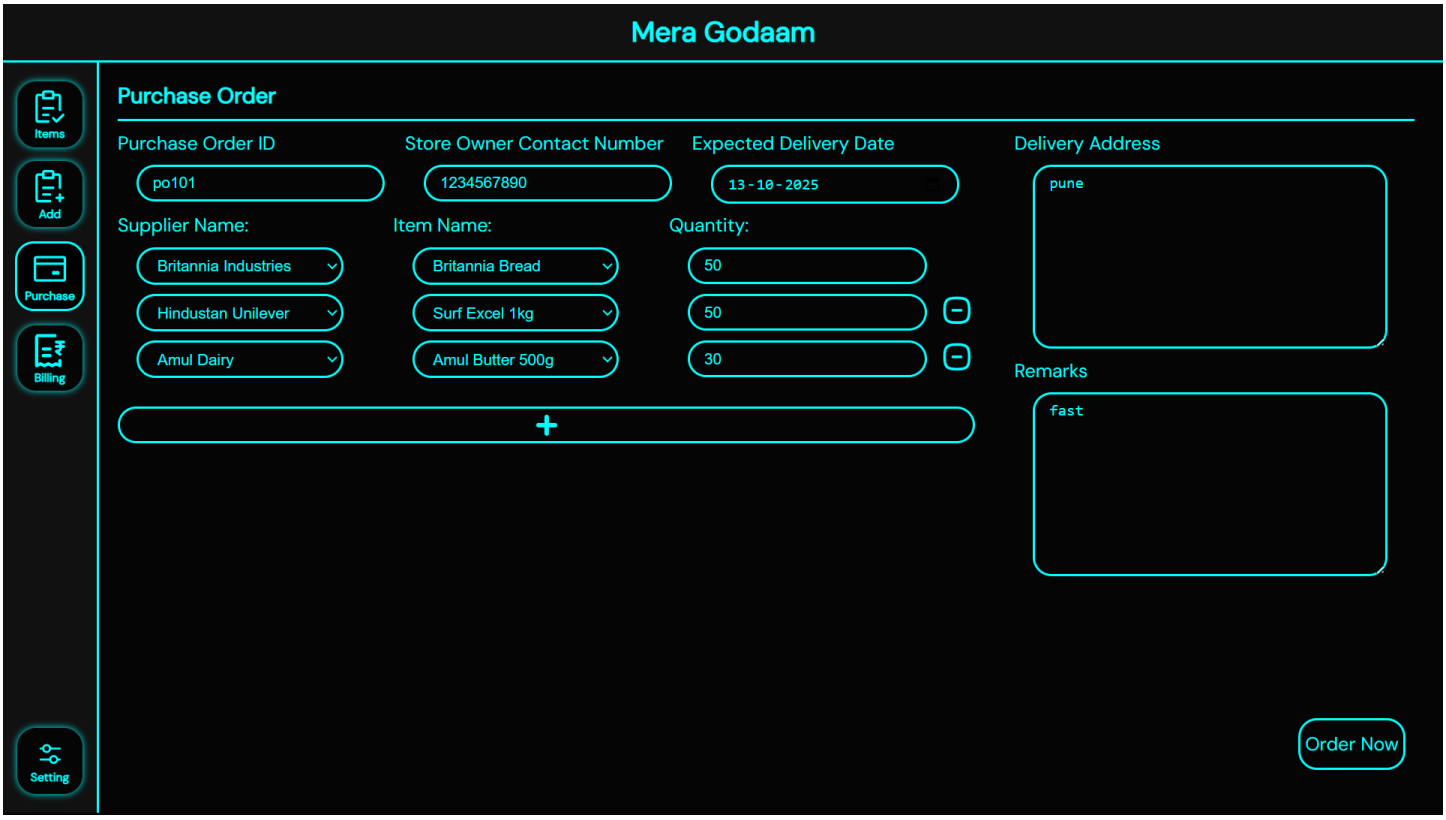
Date Added: 12-10-2025

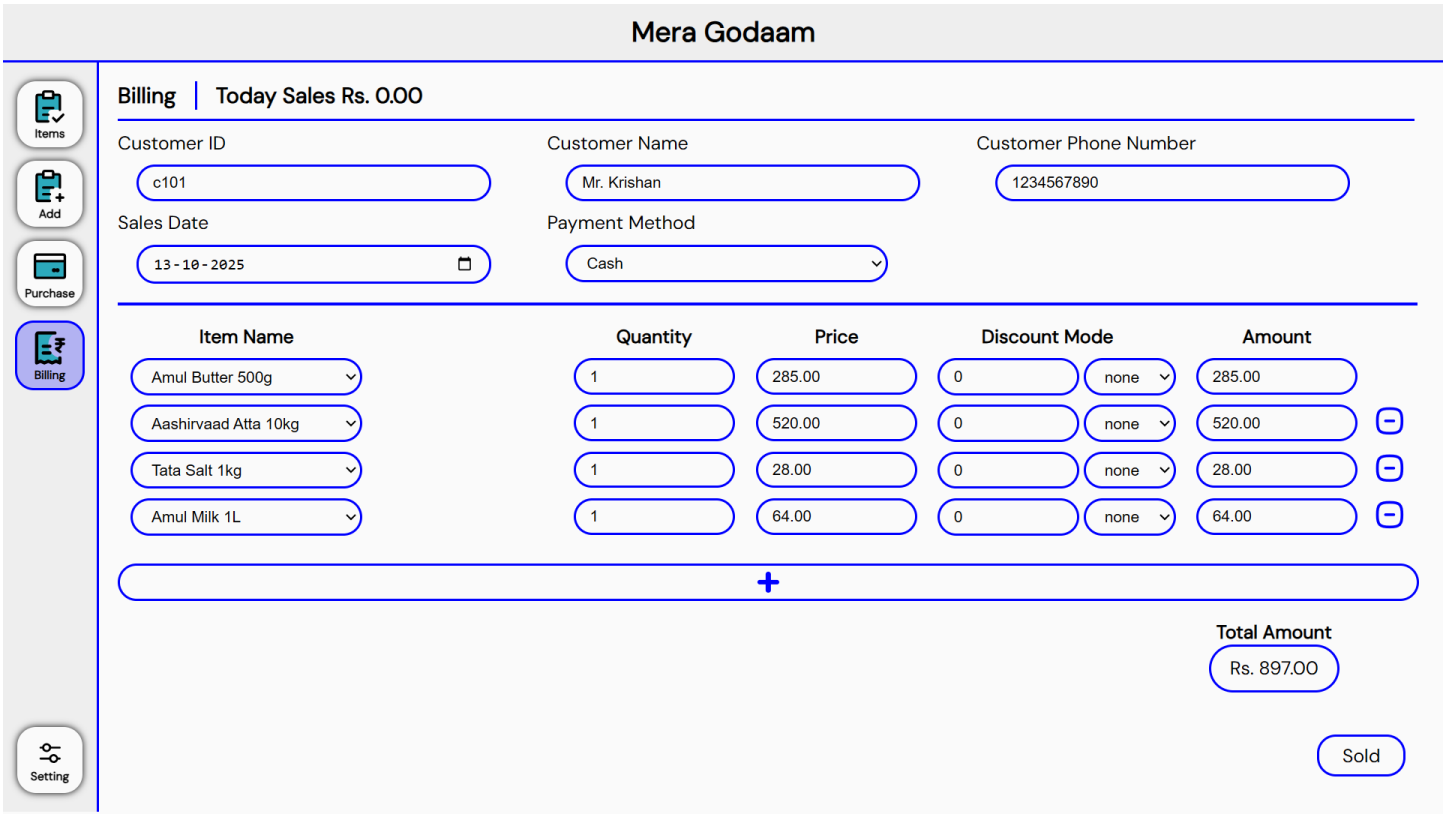


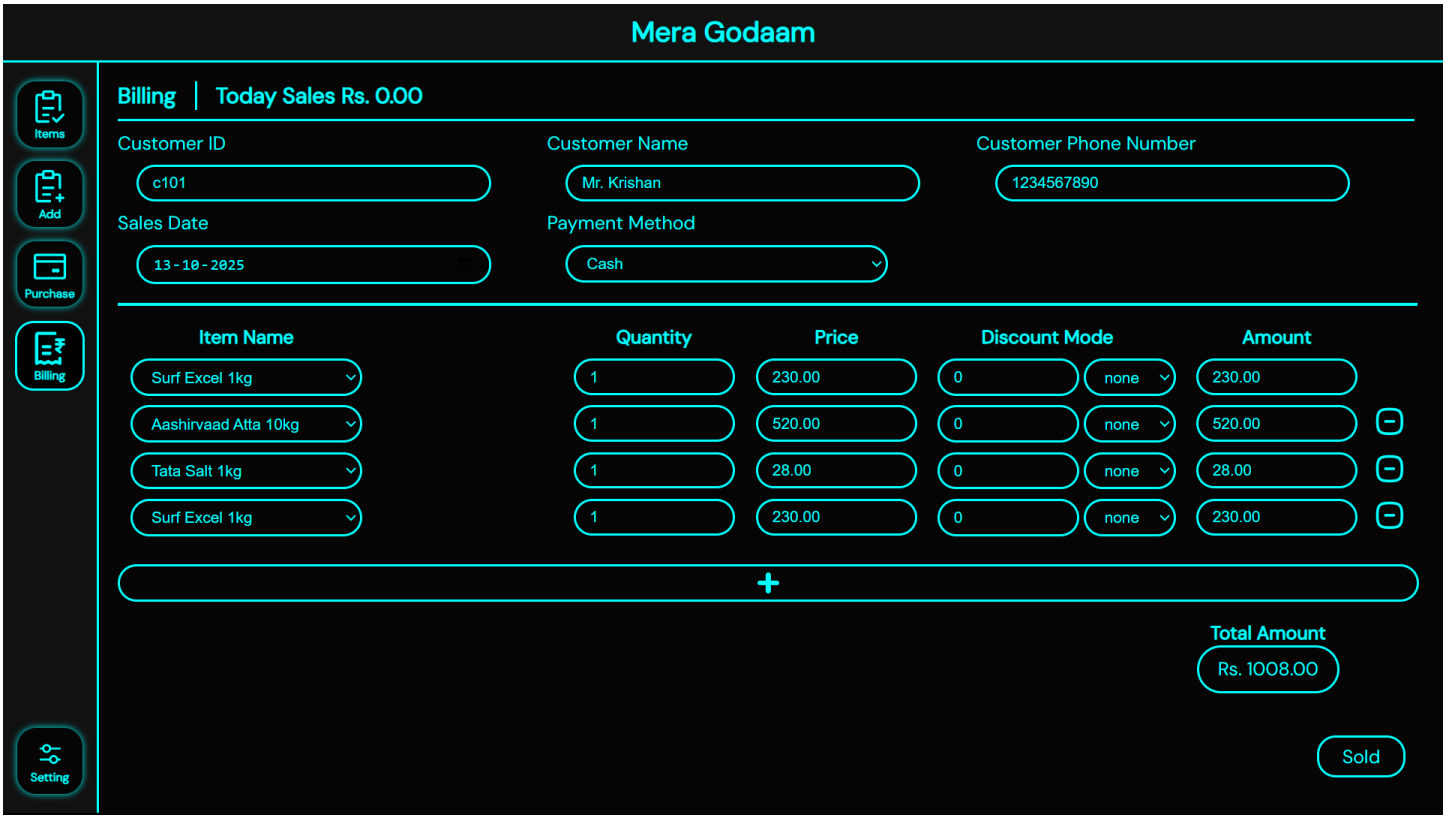
Description: Amul Milk 1L

Reset | Update | Delete


Mera Godaam
Items
Add
Purchase
Billing
Setting
Purchase Order
Purchase Order ID
po101
Store Owner Contact Number
1234567890
Expected Delivery Date
13-10-2025
Delivery Address
pune
Supplier Name:
Item Name:
Quantity:
Britannia Industries
Britannia Bread
50
Hindustan Unilever
Surf Excel 1kg
50
Amul Dairy
Amul Butter 500g
30
+
Remarks
fast
Order Now


Mera Godaam
Items
Add
Purchase
Billing
Setting
Purchase Order
Purchase Order ID
po101
Store Owner Contact Number
1234567890
Expected Delivery Date
13-10-2025
Delivery Address
pune
Supplier Name:
Item Name:
Quantity:
Britannia Industries
Britannia Bread
50
Hindustan Unilever
Surf Excel 1kg
50
Amul Dairy
Amul Butter 500g
30
+
Remarks
fast
Order Now


Mera Godaam
Items
Add
Purchase
Billing
Setting
Billing | Today Sales Rs. 0.00
Customer ID
c101
Customer Name
Mr. Krishan
Customer Phone Number
1234567890
Sales Date
13-10-2025
Payment Method
Cash
Item Name
Quantity
Price
Discount Mode
Amount
Amul Butter 500g
1
285.00
0
none
285.00
Aashirvaad Atta 10kg
1
520.00
0
none
520.00
Tata Salt 1kg
1
28.00
0
none
28.00
Amul Milk 1L
1
64.00
0
none
64.00
+
Total Amount
Rs. 897.00
Sold


Mera Godaam
Items
Add
Purchase
Billing
Setting
Billing | Today Sales Rs. 0.00
Customer ID
c101
Customer Name
Mr. Krishan
Customer Phone Number
1234567890
Sales Date
13-10-2025
Payment Method
Cash
Item Name
Quantity
Price
Discount Mode
Amount
Surf Excel 1kg
1
230.00
0
none
230.00
Aashirvaad Atta 10kg
1
520.00
0
none
520.00
Tata Salt 1kg
1
28.00
0
none
28.00
Surf Excel 1kg
1
230.00
0
none
230.00
+
Total Amount
Rs. 1008.00
Sold


Mera Godaam
Items
Add
Purchase
Billing
Setting
Settings
Help
Theme


Mera Godaam
Items
Add
Purchase
Billing
Setting
Settings
Help
Theme

❖ Conclusion

The Inventory Management System project has provided me with valuable practical experience in developing a complete web-based application that solves real-world business problems. This system successfully manages various store operations such as billing, item tracking, sales management, and purchase order, all in one single platform. It helps to maintain accurate records of stock levels, generate bills automatically, and reduce peoples effort and mistake in manual inventory management.

Through this project, I learned how to connect the front-end and back-end using technologies like HTML, CSS, JavaScript, Node.js with Express.js, and MySQL, and how to handle dynamic data efficiently. I also gained experience in designing a clean and user-friendly interface that allows smooth navigation between modules such as billing, item management, suppliers, and sales. The process of implementing CRUD operations, generating PDF bills, and calculating daily sales taught me the importance of database management and backend logic in real-world applications.

This project not only strengthened my technical knowledge but also improved my analytical and problem-solving skills. It gave me a better understanding of how software development works in a structured way from system design, database modeling, and user interface creation to testing and deployment.

Overall, the Mera Godaam: Inventory Management System is a reliable, efficient, and expandable solution that meets the needs of small and medium level stores. It has also given me a strong foundation for future projects, and with further improvements such as barcode integration, analytics, and cloud support, it can improve into a professional-level inventory management tool.

❖ References / Bibliography

AI: ChatGPT, Gemini

YouTube

MDN Web Docs

https://developer.mozilla.org/en-US/docs/Web/HTML

https://developer.mozilla.org/en-US/docs/Web/CSS

https://developer.mozilla.org/en-US/docs/Web/JavaScript